Recycling as A Sustainable Design Option

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1

Sustainability and Green Pavements

Going Green with Sustainable Asphalt Recycling, Reduction in Energy and the Carbon Footprint

2

Sustainable Pavement

- Requires less energy to construct an asphalt pavement
- Per ASCE…20% less
- Less energy consumed by public
- Less maintenance keeps traffic moving

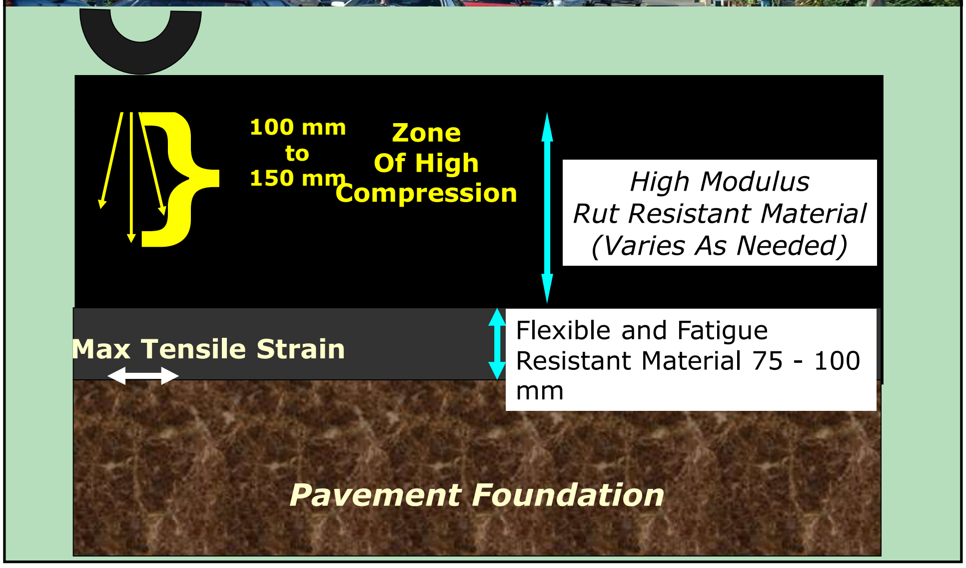
3

Sustainability in Pavement Design

- Pavements can be designed to be Perpetual Pavements
- Controlling Carbon Footprint
- Utilising Reclaimed Asphalt Pavements (RAP) in Recycling
- Warm Mix Asphalt
- Porous Asphalt
- Energy & Emission Reductions

4

Perpetual Pavements



5

Quite thick pavement structures

6

Carbon Footprint, What is it?

The total amount of greenhouse gases produced to directly and indirectly support human activities, usually expressed in equivalent tons of (CO_2)

Environmental Sustainability

7

What is a Greenhouse Gas?

- Gases that trap heat in the atmosphere
- Some occur from nature, others from human activities
  - Principal greenhouse gases entering the atmosphere from human activities are:
    - Carbon dioxide (CO_2)
    - Methane (CH_4)
    - Nitrous Oxides (N_2O)
    - Fluorinated gases (HFCs & PFCs)

8

Recyclability

- Asphalt is the No. 1 recycled material
- Biggest Re-user
  - 100 Million tons reclaimed
  - 80% recycled
- 2006 Contractor Survey
  - ~69% reused in HMA
  - ~27% other recycling
  - <3% discarded

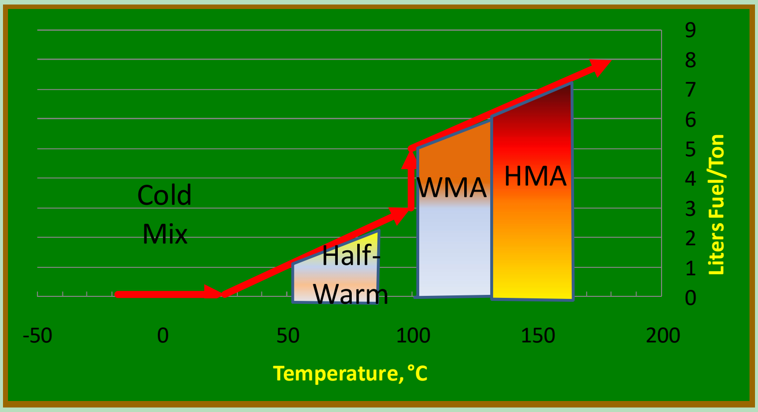
9

Why Recycle RAP into HMA?

- Best and Highest use
- Reduces demand for new materials
- Reduce demand for landfill and transporting RAP to landfill
- Reduces carbon footprint
- Contains valuable materials
  - Aggregate ~95% >$30/ton
  - Bitumen ~5.5% > $1200/ton

10

Warm Mix Classifications



11

Lower Temperature Advantages

- Lower energy consumption (≈30%)
- Lower plant wear
- Decreased binder ageing
- Early site opening – Return to user
- Cool weather paving
- Compaction aid for stiff mixes
- Cooler working conditions

12

Advantages of Recycling

- Reduced cost of construction
- Conservation of aggregate and binders
- Preservation of existing pavement geometrics
- Preservation of environment
- Conservation of energy
- Less user delay

13

Future of Recycling

- Increased reuse of materials
- Landfills becoming more and more scarce
- New equipment and admixtures versus vested interest in existing technology and equipment
- The same pavement can be recycled over and over again

14

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Recycling is one of several rehabilitation alternatives

15

Reasons that Rehabilitation is Needed

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- Inadequate ride quality
- Excessive pavement distress
- Reduced surface friction
- Excessive maintenance requirement
- Unacceptable user costs
- Inadequate structural capacity for planned use or projected traffic volumes

16

Steve Irwin, Australia

17

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18

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19

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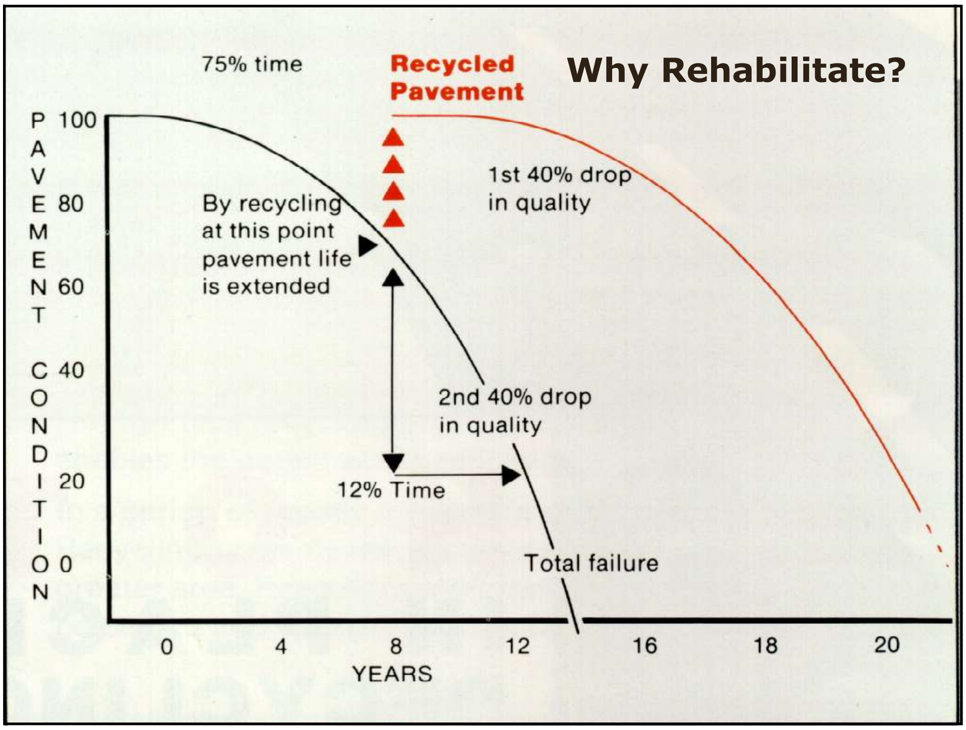
20



21

Maintenance

- Corrective – corrects or prevents deterioration from environmental effects
- Preventive – activities intended to extend or preserve the service life of a pavement, no increase in structure

22

Reconstruction

- Most extreme type of rehabilitation
- Required to improve geometrics
- Essentially delivers a “new” pavement

23

Recycling Methods

- Hot In-place Recycling
- Cold planning
- Hot Mix Recycling (central plant)
- Cold Mix Recycling (central plant)
- Cold In-place Recycling
- Full Depth Reclamation

24

Hot In-place Recycling: Process

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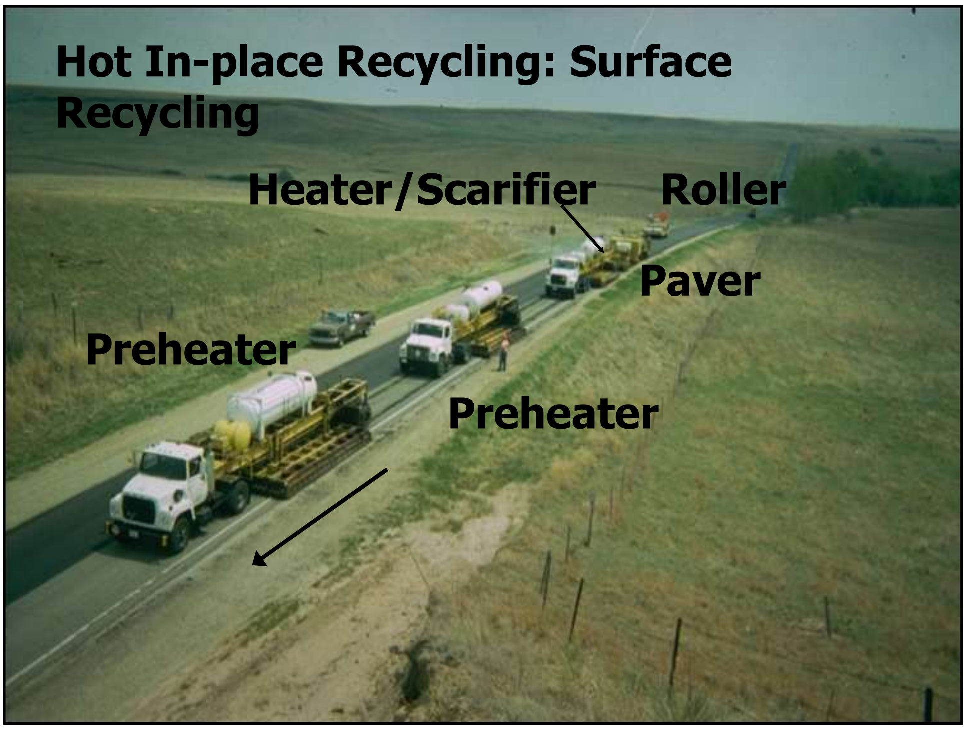
- Existing asphalt surface is heated
- Scarified to a depth from 20 to 60 mm
- Scarified material combined with aggregate, asphalt binder, and/or recycling agent
- Compaction

25



26

Hot in Place Recycling

27

28

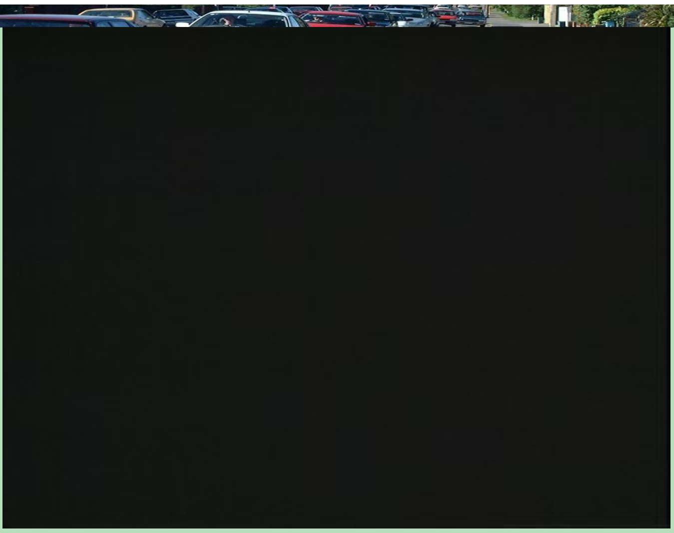

Hot In-place Recycling:
Remixing
Remixing
Machine
Virgin
Material
Recycled
Mix
Pugmill
Heater
Heater
Scarifier

29


Hot Mix Recycling in a Drum Mix Plant
Virgin
Aggregate
RAP

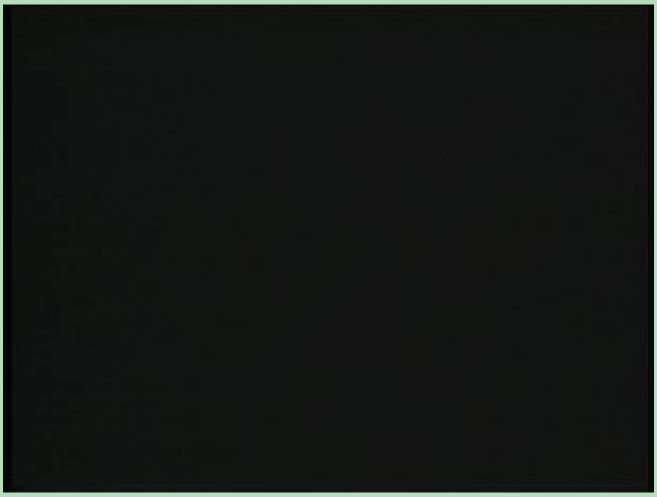
30

Hot in Plant Recycling



31

Hot in Plant Recycling



32

Cold Mix Recycling (at central plant)

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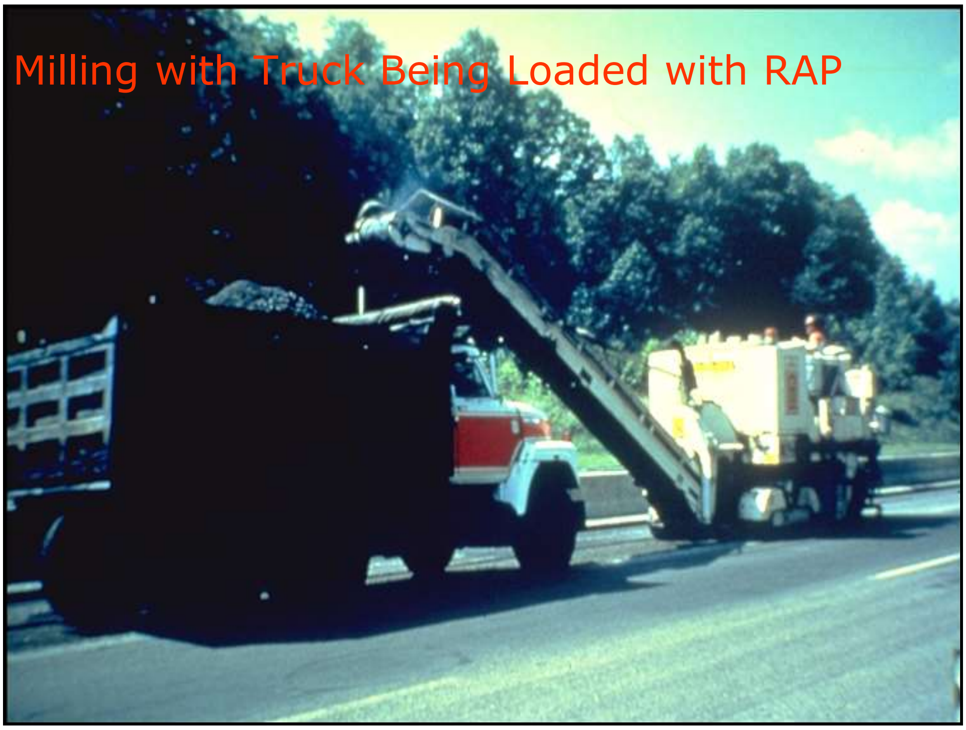
- RAP and reclaimed aggregate (if any) mixed with new asphalt binder and new aggregate (if needed) to produce cold, recycled mix without application of heat.
- Recycled mix is produced at a central plant rather than in-place.

33

Ripping Process

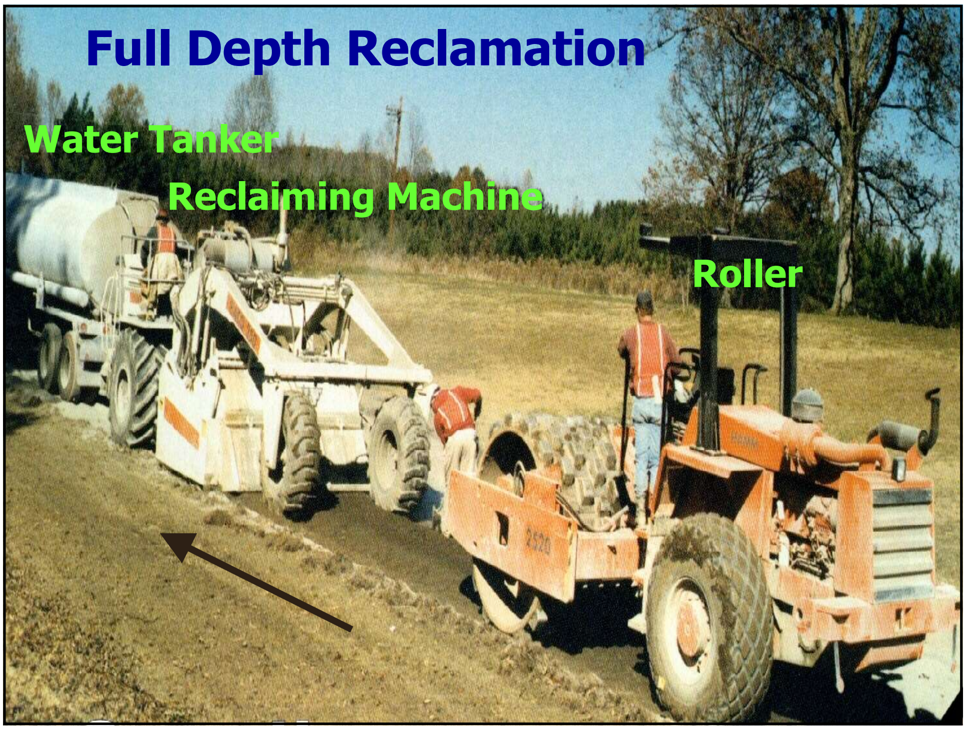
34


Milling with Truck Being Loaded with RAP

35

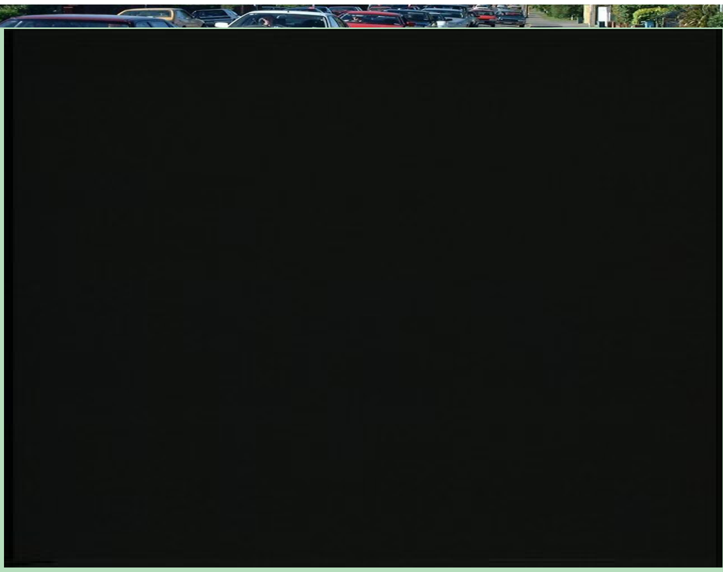

Cold In-place Recycling
Milling
Machine
Crusher
Mixer-Paver

36



37

Full Depth Reclamation



38

Project Evaluation

1. Pavement assessment
2. Historic information
3. Pavement properties
4. Distress evaluation
5. Preliminary rehabilitation selection
6. Economic analysis

39

Pavement Assessment

- Surface Deterioration
- Deformation
- Cracking

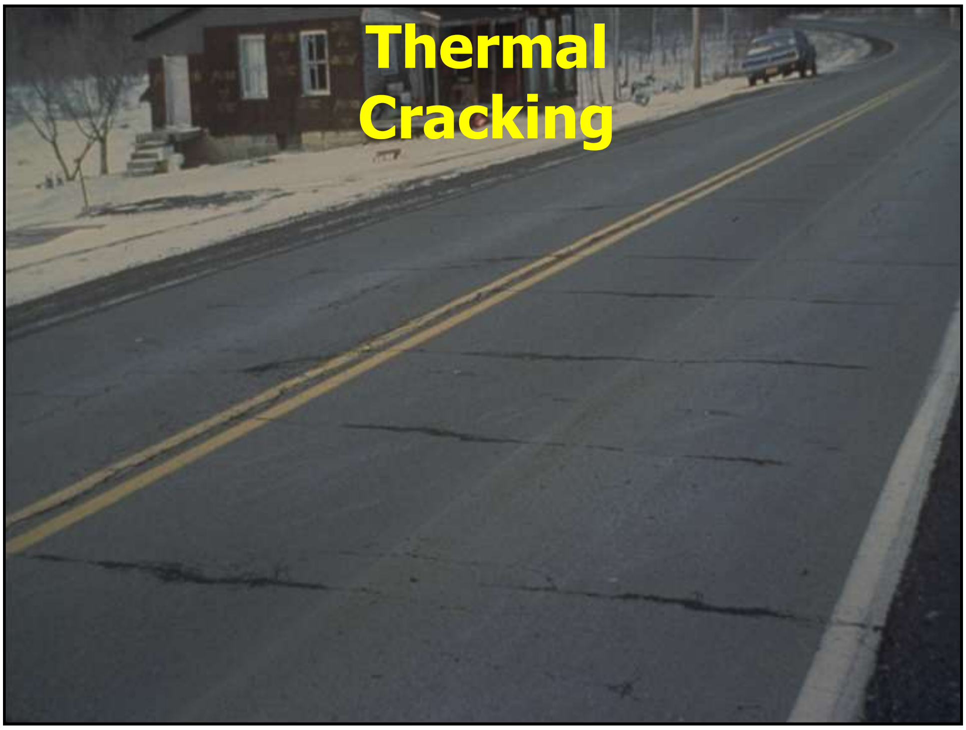
40

41

42

43

44



45

Historic Information

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- Original design
- As-built data
- QC/QA data
- PMS – pavement performance data
- Maintenance activity

46

Pavement Properties

- Roughness
- Rutting
- Friction
- Strength
- Material properties

47

Distress Evaluation

- Determine where the weakness in a pavement is located.

48

Economic Analysis

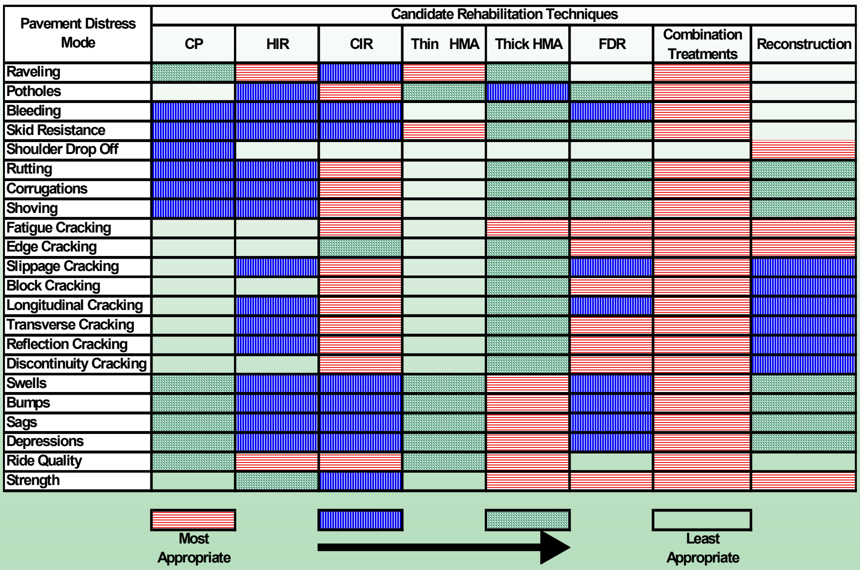
- Initial rehabilitation cost – project budget
- Future rehabilitation cost
- Maintenance cost
- Salvage value
- User costs
- Life-cycle costs
  - Present worth (PW)
  - Equivalent annual uniform cost (EAUC)

49

Rehabilitation Considerations

- Engineering considerations, e.g.
  - Type of pavement distress
  - Capacity requirements
  - Environmental impacts
  - Material availability
- Economic considerations, e.g.
  - Project budget
  - Costs vs. Benefits
  - Impacts on adjacent businesses

50



51

Any Questions?

52